

Retail Store Performance Dashboard

Compare stores on sales, footfall conversion, basket value, refunds and gross margins.

BUSINESS BRIEF | CLIENT REQUIREMENTS | DATA MODEL | STEP-BY-STEP BUILD

What the project does

Compare stores on sales, footfall conversion, basket value, refunds and gross margins. It turns the specified source tables into an interactive, filterable view of these business outcomes.

Synthetic demo checkpoints

Reference figures below describe the included synthetic dataset only; they are not client results.

REFERENCE KPI	DEMO VALUE
Net Sales	PKR 618,956,936.70
Gross Profit	PKR 235,203,635.90
Visitors	656,558
Transactions	180,977
Store Conversion	27.6%

REQUIREMENTS NOTE | This guide translates the repository specification into an implementation brief. Confirm it with the actual client; it is not a signed discovery or acceptance document.

01 / Business problem and client requirements

Agree the outcome and KPI definitions before loading customer data.

Business problem

Compare stores on sales, footfall conversion, basket value, refunds and gross margins.

Without consistent definitions, business totals can be difficult to compare or trace back to a responsible team, customer, product, or period. The dashboard is designed to let managers review the specified KPIs, apply business filters, and inspect supporting detail. This is the intended use inferred from the project specification, not a claim about a named client's current process.

Requirements interpreted from the project specification

- Provide a Power BI report that supports the stated goal: Compare stores on sales, footfall conversion, basket value, refunds and gross margins.
- Show the 3 project pages, their KPI cards, charts, filters, and detail tables listed in the page blueprint.
- Calculate the named measures using their documented formulas and preserve the project rules.
- Use the reporting scope: 180 trading days through 30 Sep 2026; one daily record per store.
- Allow analysis by: Calendar.Month, Stores.Format, Stores.Region
- Deliver editable PBIP report and semantic-model source, a data contract, private-source templates, and a reconciliation path.

Project boundaries

0 <= Transactions <= Visitors; Refunds <= GrossSales. NetCOGS must already reflect returned inventory cost. Transaction and visitor counts are events, not distinct people. Targets are one per store/date. No SKU inventory or customer-level identity is implied.

Client discovery questions

Confirm source systems and data owners; reporting currency and timezone; period and as-of date; refresh frequency; exact business definitions; privacy and access requirements; who signs off; and whether publishing to Power BI Service is in scope.

02 / Data contract and model

Keep the input tables at their documented grain; the grain determines what a summed KPI means.

TABLE	GRAIN / KEY	FIELDS (SOURCE TYPE)
Stores	one store / StoreID	StoreID (string), Store (string), Region (string), Format (string)
Trading	one store per trading date / StoreDayID	StoreDayID (string), StoreID (string), Date (dateTime), Visitors (int64), Transactions (int64), Units (int64), GrossSales (decimal), Refunds (decimal), NetCOGS (decimal), Target (decimal)
Metrics	one metadata row / Label	Label (string)
Calendar	one calendar date / Date	Date (dateTime), Year (int64), Month (string), Quarter (string)

Relationships in the PBIP model

FROM (MANY)	TO (ONE)	FILTERING
Trading.StoreID	Stores.StoreID	oneDirection
Trading.Date	Calendar.Date	oneDirection

Data handoff: Use data/import-template with exact filenames and headers. Keep client source files in approved private storage; do not commit them. Set DataMode=Folder and DataFolder to the approved location only after validating the inputs.

03 / KPI and DAX glossary

Treat these formulas and definitions as a signed calculation contract.

MEASURE	DAX SPECIFICATION	MEANING
Gross Sales	SUM(Trading[GrossSales])	Sum of GrossSales at the recorded grain.
Refunds	SUM(Trading[Refunds])	Sum of Refunds at the recorded grain.
Net Sales	[Gross Sales] - [Refunds]	Gross sales less refund amounts, excluding tax.
Net Cost of Sales	SUM(Trading[NetCOGS])	Cost of sold goods less inventory cost restored by returns.
Gross Profit	[Net Sales] - [Net Cost of Sales]	Net sales less net COGS.
Gross Margin	DIVIDE([Gross Profit], [Net Sales])	Gross profit / net sales.
Visitors	SUM(Trading[Visitors])	Sum of Visitors at the recorded grain.
Transactions	SUM(Trading[Transactions])	Sum of Transactions at the recorded grain.
Units Sold	SUM(Trading[Units])	Gross units sold; return quantities are not supplied.
Store Conversion	DIVIDE([Transactions], [Visitors])	Purchase transactions / counted store visits; not unique customer conversion.
Average Basket	DIVIDE([Net Sales], [Transactions])	Net sales / purchase transactions.
Sales Target	SUM(Trading[Target])	Daily store net sales target.
Target Attainment	DIVIDE([Net Sales], [Sales Target])	Net sales / target.
Refund Rate	DIVIDE([Refunds], [Gross Sales])	Refund amount / gross sales.

Format currency using the client's approved reporting currency. Show rate, margin, attainment, and share measures as percentages. Format units, hours, days, and counts as whole or decimal numbers only where their source supports it.

04 / Build step by step | 1-5

Work through the stages in order and preserve the original demo as your baseline.

1. Confirm the business decisions

Start with: Compare stores on sales, footfall conversion, basket value, refunds and gross margins. Confirm who will use the report and what action each page supports. Agree the reporting period, currency, refresh cadence, data owner, KPI definitions, cutoffs, and acceptance owner. The brief here is inferred from this repository spec, not a signed client requirements document.

2. Request the right source extracts

Request the tables at this grain: Stores: one store, key StoreID; Trading: one store per trading date, key StoreDayID. Use the exact file names, headers, primary keys, and types listed in the data dictionary. Ask for historical periods only where the project scope supports them; do not assume snapshots can reproduce history.

3. Keep data secure and test its quality

Place client files in approved private storage, outside GitHub. Check unique keys, required fields, valid dates, numeric signs, foreign keys, and coverage. Preserve these rules: $0 \leq \text{Transactions} \leq \text{Visitors}$; $\text{Refunds} \leq \text{GrossSales}$. NetCOGS must already reflect returned inventory cost. Transaction and visitor counts are events, not distinct people. Targets are one per store/date. No SKU inventory or customer-level identity is implied. The demo is synthetic and is for testing the model, not for client reporting.

4. Open the PBIP source project

Download and extract the full repository, then open power-bi/05-retail-performance/Dashboard.pbip in Power BI Desktop. Keep the complete folder structure. PBIP saves report and semantic-model definitions as project files that can be versioned [1].

5. Check the known-good demo baseline

Refresh in Demo mode. Review DATA-CONTRACT.md, validation/expected-demo.json, and Model.SemanticModel/DAXQueries/Reconcile.dax. Record the supplied demo totals and click through every page before replacing the source.

05 / Build step by step | 6-10

Connect private data, build the model, and test the finished report.

6. Connect the client's approved data

Copy the matching CSV templates from power-bi/05-retail-performance/data/import-template/ to the client's private source folder. In Power Query > Manage Parameters set DataMode to Folder and DataFolder to that absolute path. Keep the required Metrics row. Select Transform Data to inspect types and fix errors before loading [2].

7. Prepare fields and relationships

Use the declared data types, retain one row at the required grain, and handle blanks only as documented. Check these many-to-one links in the model: Trading.StoreID > Stores.StoreID; Trading.Date > Calendar.Date. Lookup tables should filter facts; avoid fact-to-fact joins that multiply values [3].

8. Add and reconcile measures

Use measures.dax and the KPI glossary as the calculation contract. Create explicit DAX measures with approved formats. Test results under each date and business-unit filter; verify totals against an independent calculation. Measures respond to the report filter context [4].

9. Build and review the report pages

Recreate the report blueprint below: exact page purpose, slicers, KPI cards, chart series, and detail rows. Add clear titles, sort time labels chronologically, format currency and percentages, and test interactions. Keep project-specific definitions visible to the report user.

10. Sign off, publish, and hand over

Refresh, reconcile source totals, test edge cases, document KPI ownership, and obtain business sign-off. Keep the PBIP source in version control. If service publishing is in scope, publish to the approved workspace, configure credentials and gateway if needed, schedule refresh, and test permissions [5]. Do not publish confidential data before access/security sign-off.

06 / Report blueprint and acceptance

Implement each page to support a specific decision and trace each number to source data.

PAGE / FILTERS	KPI CARDS	CHARTS / DETAIL TABLE
Store performance Filters: Month, Region, Format	Net Sales, Gross Profit, Target Attainment, Store Conversion	Charts: Net Sales by Store; Net Sales by Month Detail: Store, Net Sales, Gross Margin, Target Attainment, Store Conversion
Footfall and baskets Filters: Month, Region, Format	Visitors, Transactions, Average Basket, Units Sold	Charts: Store Conversion by Store; Transactions by Month Detail: Store, Visitors, Transactions, Store Conversion, Average Basket
Margin and refunds Filters: Month, Region, Format	Gross Sales, Refunds, Refund Rate, Gross Margin	Charts: Gross Profit by Store; Refunds by Format Detail: Store, Gross Sales, Refunds, Net Cost of Sales, Gross Profit

Acceptance checklist

- Open the PBIP in current Power BI Desktop and refresh all tables without errors.
- Reconcile source counts, key uniqueness, relationships, and every project-specific data rule.
- Compare KPI measures to independently calculated totals and the supplied expected-demo.json.
- Test each slicer, filter interaction, date behavior, chart title, table detail, blank context, and zero denominator.
- Confirm currency, percentage formats, report audiences, privacy, access, refresh ownership, and client sign-off.

Demo acceptance values

METRIC	EXPECTED (SYNTHETIC)
Net Sales	PKR 618,956,936.70
Gross Profit	PKR 235,203,635.90
Visitors	656,558
Transactions	180,977
Store Conversion	27.6%
Average Basket	PKR 3,420.09

07 / Handover and references

Use the Microsoft guidance below for product workflows; verify tenant-specific settings at delivery.

- Document approved data owner, source, refresh schedule, and support contact.
- Keep the metric dictionary, data contract, sign-off totals, and known limits with the delivered report.
- For Service deployment, confirm target workspace, credentials, gateway needs, permissions, tenant policies, and applicable license.
- A slicer does not enforce row-level security. Design and test security separately before exposing restricted data.
- This repository contains synthetic examples and a source implementation. Desktop rendering, engine evaluation, Service deployment, and client acceptance are separate release gates.

Microsoft Learn references

[1] Power BI Desktop projects (PBIP) - project format and folders

<https://learn.microsoft.com/en-us/power-bi/developer/projects/projects-overview>

[2] Connect to data in Power BI Desktop - Get Data and Power Query

<https://learn.microsoft.com/en-us/power-bi/connect-data/desktop-connect-to-data>

[3] Understand star schema and the importance for Power BI

<https://learn.microsoft.com/en-us/power-bi/guidance/star-schema>

[4] Tutorial: Create your own measures in Power BI Desktop

<https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-tutorial-create-measures>

[5] Publish semantic models and reports from Power BI Desktop

<https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-upload-desktop-files>

References checked 30 September 2026. Power BI interface labels, preview features, licensing, and tenant controls can change; confirm them in the client's environment.